engineering 4 us all

Student Handout

Name: Gadiel, Thomas, Robert, Edward **Date:** 2/7/2024 **Class:** 5

MyDesign Portfolio

Element B Initial Template

How to use this template

Students: The template below is one way to structure your research findings around past and current attempts to solve the problem, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own research into prior design attempts, be sure to delete the instructions.

Element B: Documentation and Analysis of Prior Design Attempts

Prior Design Attempt #1 - <insert name of prior design attempt #1>

Source

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some free online journals and research databases), professional organizations, government agencies, local and national newspapers, patents (here is a patent search site for the United States Patent and Trademark Office), and reputable vendors and manufacturers. Make sure to incorporate in-text citations in the details section below (i.e. don't just give a list at the end of the element or footnote).

Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

Include main details about this prior design attempt

Analysis

Name: Gadiel, Thomas, Robert, Edward **Date:** 2/7/2024 **Class:** 5

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Prior Design Attempt #2 - <insert name of prior design attempt #2>

Source

State the source of the information about this prior design attempt, which should be taken from any of the following credible source categories: peer-reviewed journals (check out your public library for free digital access to journals; here are some free online journals and research databases), professional organizations, government agencies, local and national newspapers, patents (here is a patent search site for the United States Patent and Trademark Office), and reputable vendors and manufacturers. Make sure to incorporate in-text citations in the details section below (i.e. don't just give a list at the end of the element or footnote).

Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

Include main details about this prior design attempt

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Prior Design Attempt #3 - <insert name of prior design attempt #3>

Source

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Name: Gadiel, Thomas, Robert, Edward **Date:** 2/7/2024 **Class:** 5

in-text citations in the details section below (i.e. don't just give a list at the end of the element or footnote).

Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

Include main details about this prior design attempt

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Prior Design Attempt #4 - <insert name of prior design attempt #4>

Source

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Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

Include main details about this prior design attempt

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Name: Gadiel, Thomas, Robert, Edward **Date:** 2/7/2024 **Class:** 5

Prior Design Attempt #5 - <insert name of prior design attempt #5>

Source

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Image

Include at least one image of this prior design attempt. Make sure to cite all images.

Details

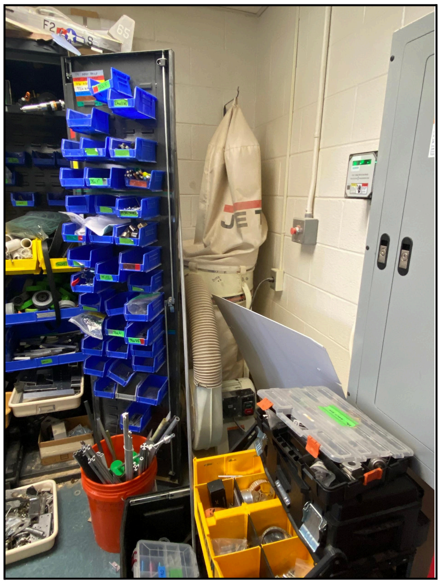
Include main details about this prior design attempt

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Name: Gadiel, Thomas, Robert, Edward **Date:** 2/7/2024 **Class:** 5

Design #1: Ventilation System



Images: [Right]

Details: Mr. L currently has in place a piping based ventilation system. The main unit is housed in the corner, restrained there due to a specialized and short cable. From this corner, piping is laid out across the walls and sides of the workshop, mounted with metal bands, and attached together “dry”, without sealant or adhesives. These pipes have 4 openings, each leading to one of the 5 woodworking tools (CNC does not have one currently), and fitted with an openable flap that blocks airflow.

Analysis: The ventilation system in place is functional and efficient enough, collecting dust from the machines and bringing it back to the JET bag in the corner. Some limitations being the “dry” fitting of the pipes, allowing some suction to escape, a possible buildup of sawdust in the piping, and students forgetting to close the flaps not in use, weakening suction and leaving dust in the air.

Name: Gadiel, Thomas, Robert, Edward **Date:** 2/7/2024 **Class:** 5

Design #2

Images:

Details: the vacuum that he has holds up to 4 gallons of dust, and has a maximum of 1 and 1/4th diameter in vacuuming. Mr. L uses this vacuum every time that someone uses the shop.

Analysis: This information was provided by Mr. L, the owner of the vacuum, and this product works well but isn't enough to get rid of all of the dust in the room.

Name: Gadiel, Thomas, Robert, Edward **Date:** 2/7/2024 **Class:** 5

Design #3:

Images:

Details: The AFS is an air conditioning system that is attached to the ceiling of a workshop, collecting dust particles and debris from the air. It is able to be controlled with a timer and a remote.

Analysis: Mr. L currently has one in place, and when it had clean filters then it did a good job at sucking in dust when the workshop was in use. The only issue is its power requirement, and the lack of clean filters to replace with. There may also be conflict with other vacuums in the shop when both are on at the same time.

Name: Gadiel, Thomas, Robert, Edward **Date:** 2/7/2024 **Class:** 5

Design #4:

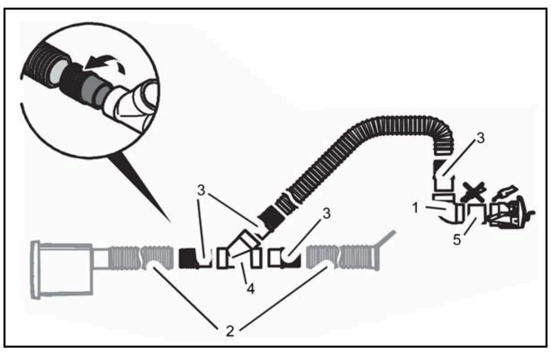
Images:

Details: This solution to the dust problems minimizes the amount of dust that students, custodians, and the teacher breathes in which has multiple health benefits as it prevents dust from going into the lungs.

Analysis: This is a temporary and easy solution to the overarching problem of dust getting into the lungs. Mr. L has not tried this before but other people did and found success with it.

Name: Gadiel, Thomas, Robert, Edward **Date:** 2/7/2024 **Class:** 5

Design #5: VacPort



Images:

Details: The VacPort is an intuitive solution, a simple vacuum cleaner mounted in the wall, attached with a hose to a pump so suction can go underway. The port is level with the ground, and can be turned on and off.

Analysis: The VacPort is a pretty fun product, as it makes sweeping debris easier; just move it towards the port so that it can be sucked up fast and quick. The issue with applying it to other areas is the solidity and difficulty of installation, especially to walls made of concrete or brick.